Resilient Cities and Structures 3 (2024) 34–45

Contents lists available at ScienceDirect

Resilient Cities and Structures

journal homepage: www.elsevier.com/locate/rcns

Full Length Article

# Digital twin approach for enhancing urban resilience: A cycle between virtual space and the real world

Yixing Wang [a], Qingrui Yue [b], Xinzheng Lu [c], Donglian Gu [b,*], Zhen Xu [b], Yuan Tian [b], Shen Zhang [d]

[a] Institute of Engineering Mechanics, China Earthquake Administration, Harbin 150000, China
[b] Research Institute of Urbanization and Urban Safety, School of Civil and Resource Engineering, University of Science and Technology Beijing, Beijing 100083, China
[c] Key Laboratory of Civil Engineering Safety and Durability of China Education Ministry, Department of Civil Engineering, Tsinghua University, Beijing 100084, China
[d] Central South Architectural Design Institute Co., Ltd., Hubei 430014, China

ARTICLE INFO

*Keywords:*
Digital twins
Disaster simulation
Resilience
Disaster prevention and mitigation

ABSTRACT

Construction of disaster-resilient cities has attracted considerable attention. However, traditional methods of studying urban disaster resilience through experimental approaches are often constrained by various limitations, such as testing sites, costs and ethical considerations. To address these constraints, this paper proposes incorporating digital twin concepts into urban disaster resilience research. By establishing a connection between the physical realm of the city and its virtual counterpart, this approach utilizes digital simulations to overcome the limitations of experimental methods and enables dynamic deduction and control of the disaster process. This paper delves into three key aspects encompassing the acquisition of data from reality to the virtual space, disaster simulation within the virtual space, and translation of virtual insights into effective disaster prevention strategies in reality. It provides a comprehensive summary of relevant research endeavors from the authors' research group and showcases the effectiveness and potential of the proposed techniques. These findings serve as references for pre-disaster planning, real-time emergency assessments, post-disaster rescue operations, and accident investigations for buildings and cities.

## 1. Introduction

"Sustainable Cities and Communities" (also known as SDG 11) is at the heart of achieving all 17 of the United Nations Sustainable Development Goals (SDGs) [1,2], aiming to ensure inclusive, safe, resilient, and sustainable cities and communities by providing affordable transportation solutions, curbing urban sprawl, promoting active participation in urban governance, safeguarding cultural assets, and addressing climate change issues. Earthquake, fire, and wind disasters are three of the most devastating disasters, resulting in immediate loss of life and long-term economic damage [3–5]. Taking China as an example, according to data released by the National Bureau of Statistics, the direct economic losses caused by earthquake, fire, and wind disasters in 2021 were 10.6, 6.7, and 13.2 billion, respectively. Therefore, enhancing the resilience of cities and communities against these disasters is of paramount importance for advancing SDG 11.

In previous studies, physical-object-based experimental methods have been widely used to test the performance and resilience levels of infrastructures, such as buildings and bridges, at an individual object scale [6–8]. Owing to limitations resulting from testing sites and costs, this method faces significant difficulties when applied at an urban scale. Hence, there is an urgent need for a more feasible, efficient, and accurate method for studying the disaster resilience of buildings and cities.

Digital twins have become popular research topics in recent years [9–11]. The earliest concept of digital twins was proposed by Grieves [12] in 2003. In 2011, the U.S. National Aeronautics and Space Administration (NASA) first introduced the concept of "twins" in the Apollo program [13] and provided a detailed definition of digital twins [14]. It refers to mapping real-world entities into the virtual space, synchronizing the virtual model with the real state of the entities, and achieving interaction between the virtual and real worlds [15]. Digital twin technology can be applied to guide decision-making in the real world through simulations in the virtual space. Hence, in the last 10 years, researchers from different fields have explored the applications of digital twins in various areas, such as product design [16–19], manufacturing [19–21], and operation condition monitoring [22–24].

In civil engineering, digital twin technology can not only be used for three-dimensional(3D) reconstruction and visualization of individual infrastructure objects[25–28] or cities[29–30], but also can be applied to disaster risk assessment and other fields[31–38].Regarding individual infrastructure objects, Ariyachandra and Brilakis [25] used a car-mounted laser scanner to obtain the coordinate positions of rail-

* Correspondence author.
*E-mail address:* gudonglian@ustb.edu.cn (D. Gu).

https://doi.org/10.1016/j.rcns.2024.06.002
Received 4 December 2023; Received in revised form 7 April 2024; Accepted 2 June 2024
Available online 11 June 2024
2772-7416/© 2024 The Author(s). Published by Elsevier B.V. on behalf of College of Civil Engineering, Tongji University. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/)

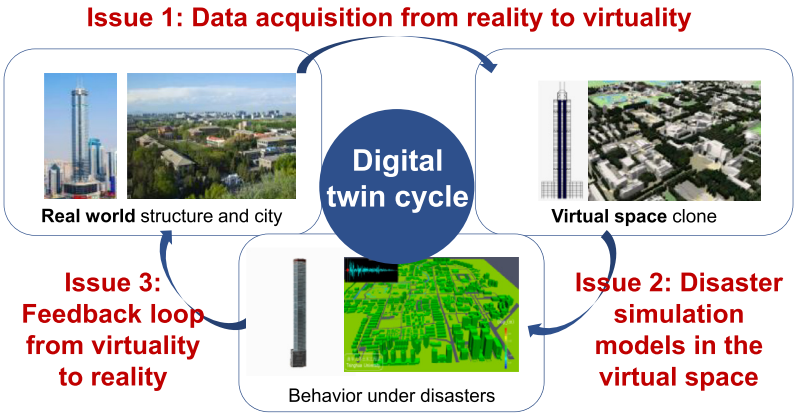



**Fig. 1.** Basic concept of digital twin-driven analysis of disaster resilience for buildings and cities.

way masts and created a building information model (BIM) for all masts along a railway line. Xiong et al. [26] employed semantic segmentation to classify point-cloud data obtained from laser scanning, generating 3D models of non-structural components within buildings. Based on autonomous drone photography, Fiorillo et al. [27] deployed entire stations at feature points within an area to acquire coordinate information and unify the coordinate data of image information. They reconstructed 3D information of the Paestum Temple in Italy. Wu et al. [28] used a smartphone-based imaging system to capture the image data of physical objects from the ground, compensating for the loss of geometric information caused by the mutual occlusion of buildings during drone photography. Based on this, they established detailed 3D models of buildings using automatic feature point matching algorithms. With respect to the city scale, many countries have been gradually carrying out the construction of digital twin cities. For example, in 2015, the Singaporean government introduced the "Virtual Singapore" project, which utilized Dassault Systemes' 3D EXPERIENCE platform to digitally model Singapore's buildings, infrastructure, trees, etc. It incorporates images and data collected by sensors for analysis, and provides urban environment simulation, operational analysis, planning, and management decision-making for residents, businesses, government agencies, and research communities. The government of New South Wales in Australia initiated a virtual four-dimensional model of the Western Sydney region, encompassing buildings, subsurface plans, terrain, property boundaries, and utilities such as electricity, water, and sewage. Taking the West Cambridge area of the University of Cambridge as an example, Lu et al. [29] proposed a digital twin system architecture for building clusters and city scales by integrating heterogeneous data sources to support historical data queries and analysis. In addition, they discussed the challenges of constructing a digital twin city from a technical perspective, based on the case of the University of Cambridge. White et al. [30] released a digital twin model of Dublin's port area, allowing citizens to report and mark city issues through interactive devices and enabling simulations of flood propagation, among other functionalities.

Moreover, digital twin technology is being investigated to assess the damage to buildings and the resilience of virtual communities. Enderami et al. [31,32] proposed a novel computational model and quantitative assessment framework that can validate and confirm the impacts, damages, and recovery processes of community-level disasters. Ellingwood et al. [33] established a virtual testbed to develop measurement science for community resilience assessment. Deierlein et al. [34,35] developed a computational workflow for regional hazard simulation, addressing the characterization and simulation of the destructive impacts of natural hazards on buildings and civil infrastructure. They also quantified the resulting economic losses, disruptions, and other societal consequences. Braik & Koliou [36] introduced a novel digital twin framework for electric power networks, which is capable of modeling and estimating faults and performance states of various components within electric power networks. Wang et al. [37] established an urban building inventory using deep learning techniques, providing the necessary data for disaster and risk management planning and simulations. Attary et al. [38] proposed a combination of electric power networks and building damage probabilities under wind disasters using techniques such as weighted cellular automata.

Although digital twin technology has been extensively applied in civil engineering, previous studies have primarily focused on the daily operation and maintenance of infrastructure objects and cities. In addition, there are also studies on urban resilience through digital twins, but this is still its infancy early stages. Further research is needed to investigate the application of digital twin technology in enhancing urban disaster resilience.

Motivated by the encouraging advancements in digital twins, introducing the concept of digital twins into research on city disaster prevention can provide quantitative guidance for promoting city resilience by predicting or reproducing the impact of disasters in the virtual space. As shown in Fig. 1, three important issues must be addressed in the digital twin-driven analysis of disaster resilience for buildings and cities.

(1) How can building and city information be efficiently obtained from the real world and mapped from reality to virtuality?
(2) How can disaster simulation models be established for buildings and cities in the virtual space to predict or simulate disaster processes accurately?
(3) How can simulation results from the virtual space be used to guide disaster prevention and mitigation strategies in the real world and achieve a feedback loop from virtuality to reality?

A digital twin cycle can be effectively established by addressing these three issues, providing a solid framework for the formation of rational disaster prevention and mitigation strategies. Based on the authors' recent research outcomes, this paper discusses these three issues. Specifically, three aspects, namely disaster simulation models for buildings and cities in the virtual space, artificial intelligence (AI)-powered building and city data acquisition from reality to virtuality, and disaster prevention and mitigation implementations based on the feedback loop from virtuality to reality, are introduced in the following sections. The research outcomes highlight the potential of digital twins in facilitating the improvement of city resilience and the achievement of SDG 11.

## 2. Disaster simulation models for buildings and cities in the virtual space

Owing to the low-frequency nature of disasters and the rapid development of cities, there is a lack of disaster prevention and mitigation




(a) Earthquake-induced collapse simulation of a super high-rise building

(b) Fire simulation for a subway station

(c) Wind field simulation around a building

**Fig. 2.** Individual-building-level damage simulation model.

experience for structures and urban areas built in recent years. Consequently, the reliability and applicability of the data-driven disaster simulation models generated by fitting historical disaster data require further evaluation. In comparison, simulation models established based on the physical laws of the disaster process exhibit favorable rationality and universality and can be applied to different regions and disaster scenarios without dependence on the availability of historical disaster data. Therefore, to ensure the rationality and reliability of disaster simulation results, it is necessary to begin with the disaster mechanism and establish a model that can truly reveal the physical laws of disaster evolution. Several physics-based damage simulation models have been developed at both the individual building and city scales by the authors, as introduced below.

### *2.1. Individual-building-level damage simulation models*

(1) Models for earthquake

Super high-rise buildings have been rapidly built in China, with little experience of seismic damage. To address their seismic vulnerability issues, Lu et al. [39] proposed a novel seismic collapse simulation model powered by a fiber beam, layered shell, and elemental deactivation technique, which can capture strong nonlinear structural dynamic features during the entire process of earthquake-induced building collapse. This model has been applied to many landmark buildings, including the tallest Chinese building (i.e., the Shanghai Center with a height of 632 m) and Beijing's tallest building (i.e., the CITIC Tower with a height of 528 m, Fig. 2a).

(2) Models for fire

Fire simulation in buildings is a sophisticated problem that interacts with both thermodynamics and fluid dynamics. Lu et al. [40] and Xu et al. [41] proposed a high-fidelity scenario construction method for fire spread and a volume-rendering-based smoke visualization method that can predict or reproduce a fire and smoke spread scene. These methods have been applied to fire simulations of several important facilities such as schools, hospitals, and subway stations (Fig. 2b).

(3) Models for wind

To achieve a high-efficiency establishment of wind simulation models, Lu et al. [42] developed an efficient mapping method from a BIM to a computational fluid dynamics (CFD) model (Fig. 2c). This method automatically extracts building façade data, based on which the established CFD model can subsequently be used to predict the wind field around the building.

### *2.2. City-scale building damage simulation models*

(1) Models for earthquake

The finite element method has been widely used in previous studies to predict the seismic performance of individual buildings [39,43]. Nevertheless, this method cannot be applied at the city scale because of the significant amount of manpower and computing resources required to perform a detailed finite element analysis of all the buildings in an urban area. To address this issue, Lu and Guan [44] developed a city-scale time history analysis (THA) method, realizing the rational prediction of the story-level seismic dynamic response of building clusters in a large urban area and achieving efficient and accurate seismic damage prediction and high-fidelity visualization of earthquake scenarios. This method, based on the basic equations of structural dynamics theory, uses multiple-degree-of-freedom (MDOF) models to simulate the structural behaviors of different types of buildings. Specifically, the MDOF shear model is used to simulate a structure whose deformation mode is dominated by shear deformation (for example, a multistory frame structure), whereas the MDOF flexural–shear model is adopted for structures with large height-to-width ratios and flexural–shear coupling deformation modes (for example, a high-rise frame–shear wall structure). In addition, a GPU/CPU collaborative computing technique is used to facilitate computational efficiency. In the absence of design drawings, an automated model generation algorithm was proposed to estimate the parameters based on building inventory data [44]. The proposed parameter determination method, based on a simulated design procedure and statistics obtained from extensive experimental and analytical studies, can be easily applied to different regions with different design requirements. Given the above-mentioned advantages, this method has been adopted by well-known international projects, including the US-NSF NHERI SimCenter, and has been applied to disaster analyses of more than 50 cities worldwide, including the San Francisco Bay Area (1.84 million buildings). By dynamically visualizing the simulation results, the displacement response of buildings is obtained (Fig. 3).

(2) Models for fire

To realistically reproduce the fire-spread scene among building clusters, Xu et al. [45] and Lu et al. [46] proposed a fire-spread simulation method based on thermal radiation and plume mechanisms and developed a smoke diffusion simulation method based on fluid dynamics. This method can not only clearly illustrate the dynamic evolution of fire among building clusters, but can also be combined with the city-scale THA method to simulate fire following earthquakes (Fig. 4b). The model considers the impact of various ground motions and the seismic resilience of different buildings on the fire spread process. The proposed method has been successfully applied in simulating fires in two villages,

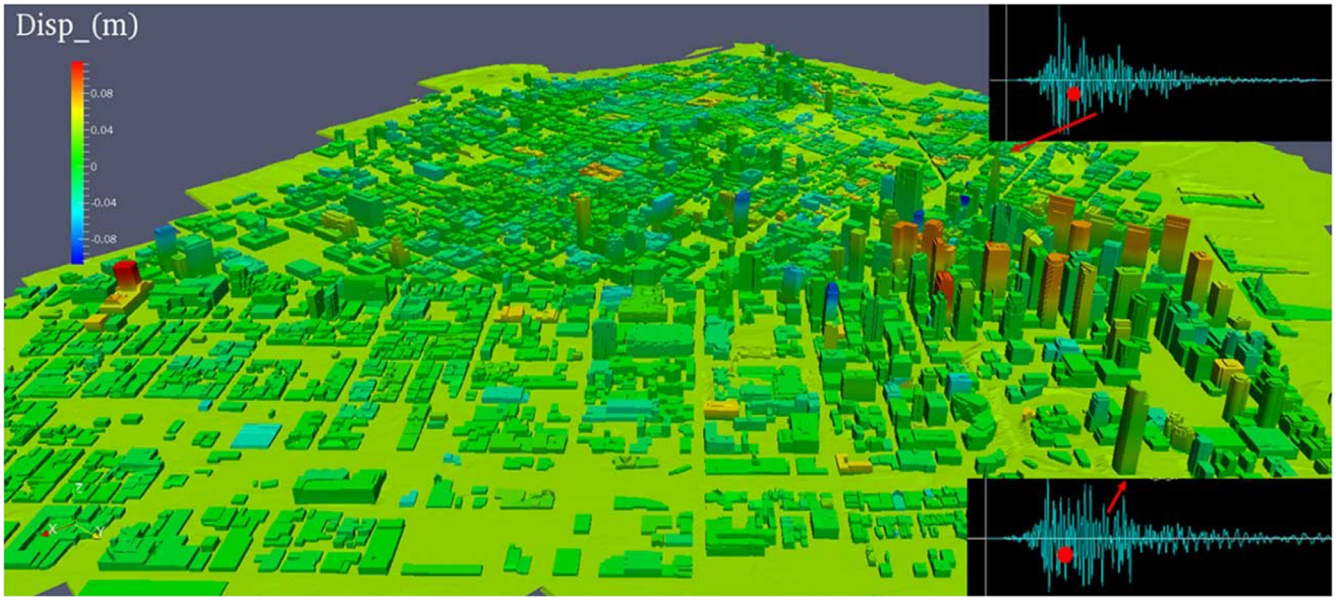
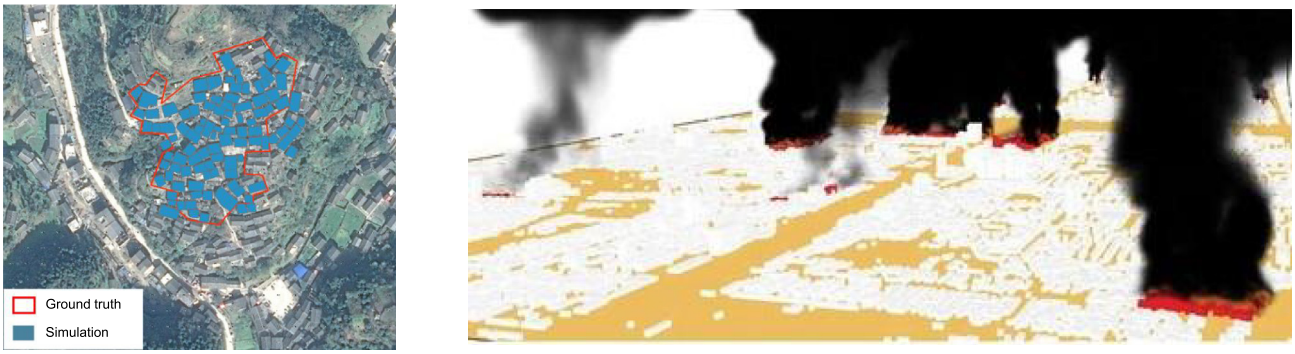



**Fig. 3.** Seismic dynamic response simulation of buildings in the San Francisco Bay Area.




(a) Fire simulation of a village in southwestern China

(b) Simulation of fire following an earthquake in a provincial capital city in China

**Fig. 4.** Applications of regional fire simulation models.

replicating the actual spread scene of two fires. The three-dimensional fire simulation closely aligns with the actual burned area of the fire (Fig. 4a). Moreover, it is evident that the motions of the smoke particles can enhance the reality of the fire scene, from which the fire location and damage severity of the urban region can be clearly identified (Fig. 4b).

(3) Models for wind

To accurately predict wind damage to building clusters, Gu et al. [47] established a novel wind damage model based on a mechanical model of building façade components and CFD, in which the scenario bank concept was introduced to facilitate the computational efficiency of CFD simulations. In addition, the influence of trees on wind pressures acting on building façade components was considered in the proposed method by modifying the Navier–Stokes equations within the tree domains. The method features an obvious advantage over regression-based empirical models in terms of accuracy and universality, and the prediction results are in good agreement with actual wind damage records.

### *2.3. AI-powered damage simulation model*

Physics-based models generally need to address a large number of physical equations, which presents a bottleneck in terms of computational efficiency. The introduction of AI may compensate for this deficiency. Taking a regional earthquake simulation as an example, Xu et al. [48] and Lu et al. [49] combined deep learning with city-scale THA and proposed a real-time seismic damage simulation method. Specifically, as shown in Fig. 5, the model comprises four main parts: historical ground motion gathering, region-specific database generation via city-scale THA, long short-term memory (LSTM) neural network training and verification, and real-time damage prediction, which establishes high-performance mapping rules between ground motion and structural damage. The computational accuracy and efficiency of this model have been validated based on a case study of a typical large community. The results indicate that the average error can be decreased by 61 % compared with the traditional vulnerability method, and the exhausted computation time can be reduced from 12 min to 0.52 s compared with the physics-based model (i.e., city-scale THA), highlighting the great potential of AI in damage simulation models.

## 3. AI-powered building and city data acquisition from reality to virtuality

Physics-based models require various types of building and city data as inputs. Consequently, efficient data acquisition from reality

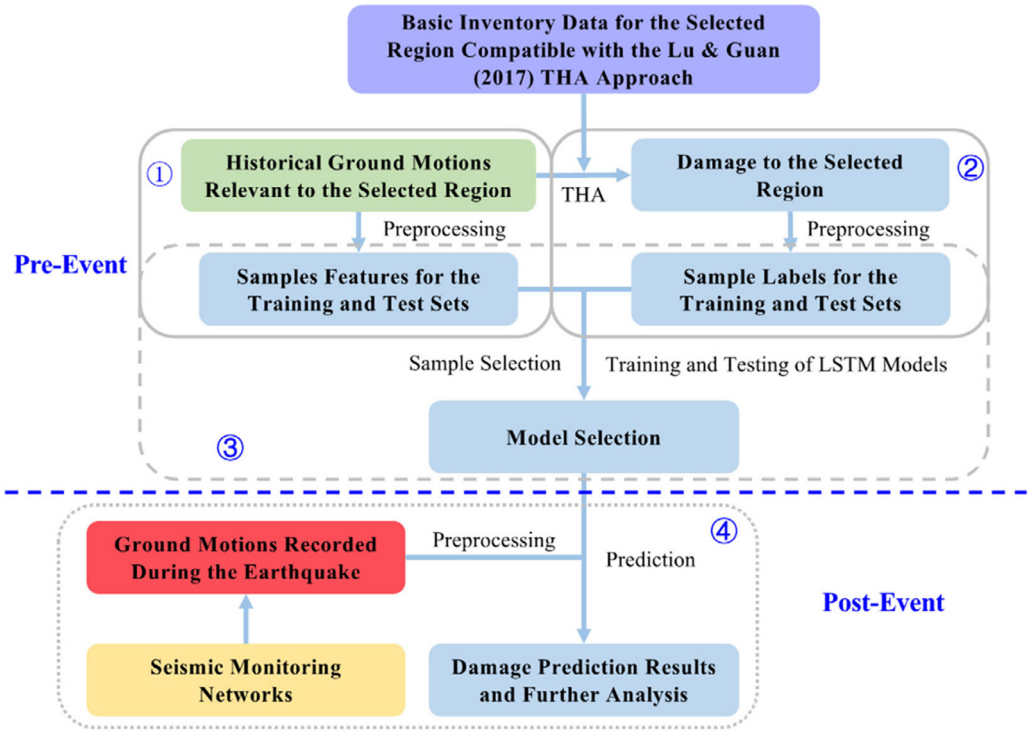



**Fig. 5.** AI-powered regional seismic damage simulation model.

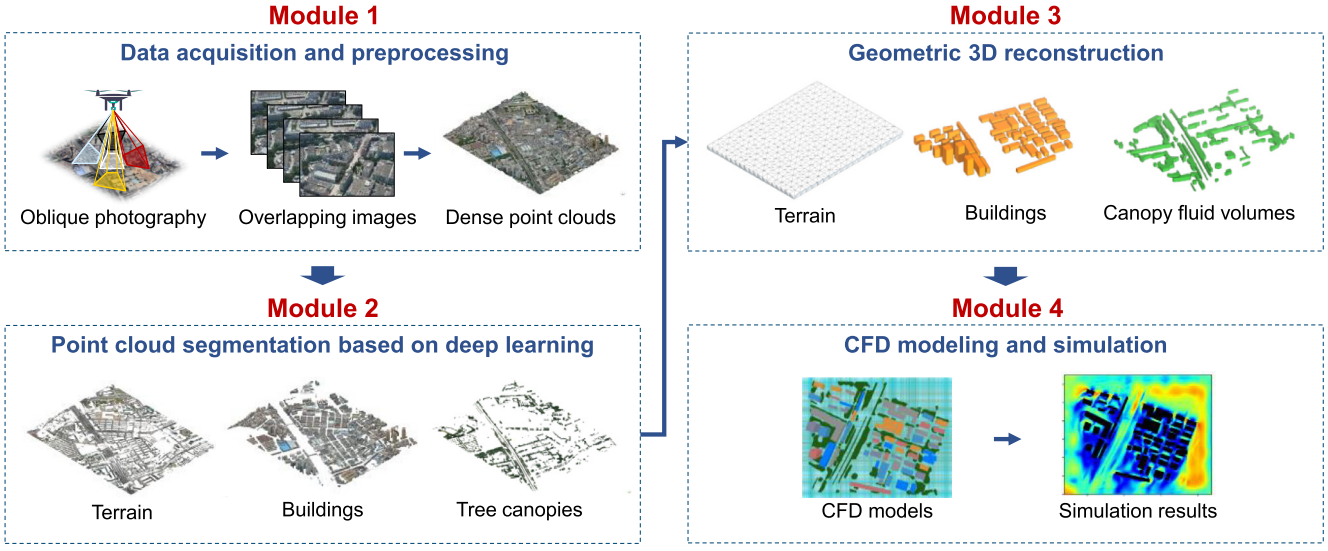



**Fig. 6.** Pre-disaster building and city data acquisition based on oblique photography.

to virtuality is important for establishing disaster simulation models and improving their accuracy. This section discusses the building and city data acquisition in three stages: before, during, and after disasters.

In previous studies, pre-disaster data acquisition was generally based on geographic information systems (GIS), the applicability of which is restricted by a low frequency of data updates and strong dependence on manual methods. Data acquisition during disasters is often based on pre-arranged sensing devices with limitations such as high cost, short lifespan, and the inability to be deployed in a large number of buildings in cities. Post-disaster data acquisition via manual field surveys is generally of low efficiency and high risk. AI was introduced to address these challenges and achieve highly efficient and accurate data acquisition in the three aforementioned stages.

### *3.1. Before disasters: building and city data acquisition based on oblique photography and street view*

Drones can be used to obtain city-scale building information quickly, exhibiting the advantage of cost efficiency compared with field investigation methods. Sun et al. [50] and Gu et al. [51] proposed a novel method to acquire building and city information using drones and deep learning. The method applies drones to obtain oblique photography data of the target area and realizes automatic information extraction of buildings, façade components (such as windows), and other objects (such as trees) through deep learning algorithms. Two-dimensional (2D) data extracted from 2D images can be transformed into 3D data in the real world based on the proposed mapping method and the drone position data recorded during flight. Consequently, the automatic establishment

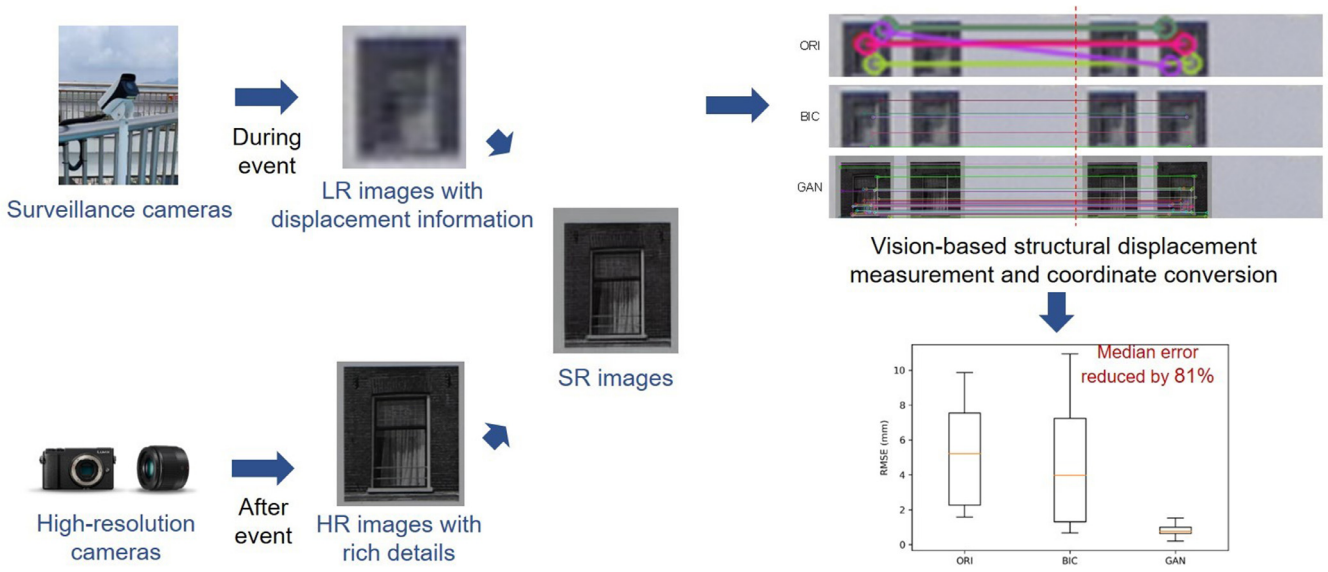



**Fig. 7.** Super-resolution method based on generative adversarial network to improve displacement measurement accuracy.

of disaster simulation models can be achieved, significantly improving the model construction efficiency compared to time-consuming and labor-intensive manual methods (Fig. 6).

### 3.2. During disasters: structural deformation data acquisition driven by computer vision

Monitoring the deformation or displacement response of buildings during disasters is critical for ensuring structural safety. The recent development of computer vision has led to extensive research on the application of vision-based measurements in structural monitoring. Urban surveillance video cameras, which are widely installed in cities, can produce numerous images and videos of urban scenes to measure structural displacement. Provided that the building response can be monitored in real-time with urban surveillance video cameras, the city-scale structural safety issue can be addressed in a cost-efficient manner without installing sensors on a large number of buildings in the city.

However, structural displacement measurements may be inaccurate owing to the limited hardware resolution of surveillance video cameras or the long distance from the cameras to the monitored targets. To this end, Sun et al. [52] proposed a novel method to improve displacement measurement accuracy using a deep learning super-resolution model based on generative adversarial networks (Fig. 7). The proposed method enhances the texture detail of low-resolution images or videos by supplementing high-resolution photographs of the target, thereby improving the accuracy of vision-based displacement measurements. The proposed method showed good accuracy and stability in both static and dynamic experimental validations compared with the original low-resolution images/video and interpolation-based super-resolution images/video. In conclusion, the proposed method can support displacement measurement of buildings and infrastructure based on urban surveillance video cameras.

Furthermore, the 3D displacement of structures is important for structural analysis. It is challenging for existing vision-based measurement methods to obtain all the 3D displacement components because they require either multi-view camera systems or additional specially designed targets, making it difficult to meet the requirements of measurement applications based on urban surveillance cameras. Sun et al. [53] proposed a 3D structural displacement measurement method using monocular vision and deep-learning-based pose estimation. This method uses virtual rendering to synthesize the training set based on 3D models of the target objects and then trains the deep learning model Dense Pose Object Detector (DPOD) to estimate the poses of the target object. Finally, it measures the 3D translation of the structures based on the original and destination poses or the original pose and keypoint matching (Fig. 8). The effectiveness of the proposed method was validated through static and dynamic experiments, indicating that the method meets the requirements for obtaining 3D structural displacement and has good accuracy in identifying the principal frequencies of the dynamic responses. The proposed method can support 3D displacement measurements of buildings and infrastructure based on urban surveillance cameras.

### 3.3. After disaster: building seismic damage data acquisition based on drones

Rapid assessment of seismic damage to buildings can facilitate improved emergency responses and timely relief in earthquake-prone areas. With the rapid development of remote sensing technology, satellite or aerial images of disaster areas have become available within 24 h after an earthquake. Collapsed buildings can be easily identified from these images. Hence, taking advantage of the identified building collapse scene in a disaster area, Xiong et al. [54] and Lu et al. [55] proposed a near real-time seismic loss estimation method for regional buildings, improving the accuracy of nonlinear THA-based loss estimations (Fig. 9). Specifically, a building image segmentation method was proposed using a 3D building model of the target area as a georeference, through which multiview-segmented building images can be obtained. Subsequently, the VGGNet convolutional neural network, which is fine-tuned based on manually tagged building images obtained from the Internet, is adopted to identify the collapse distribution of buildings. Next, a series of THA is performed for the target regional buildings, thereby generating numerous simulation results. Through a machine-learning-powered similarity measurement method, the simulation results that bear strong similarities to the identified collapse scene are identified as optimal solutions, which are used to estimate the seismic loss. The simulation results of real-world case studies indicate that the use of the identified building collapse scene leads to considerably closer estimations of the actual economic losses.

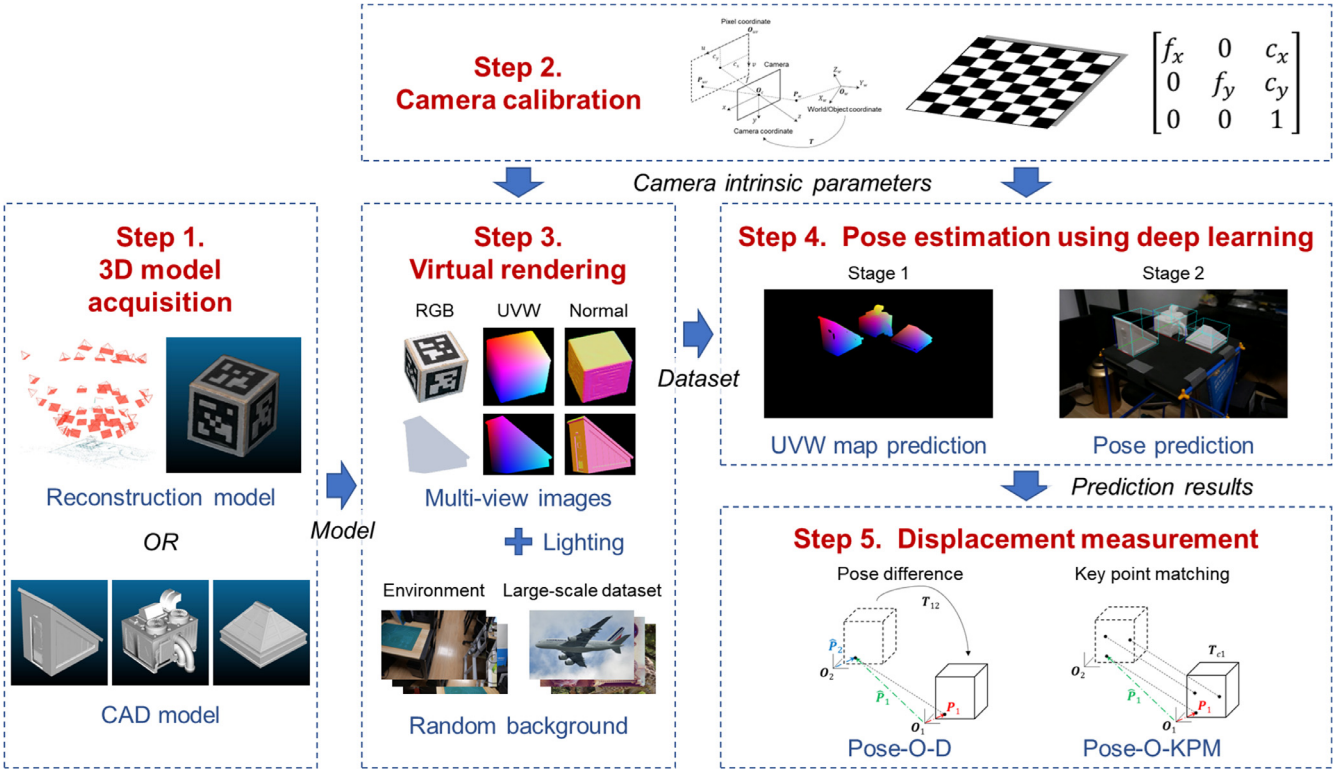



**Fig. 8.** 3D structural displacement measurement method using monocular vision and deep-learning-based pose estimation.

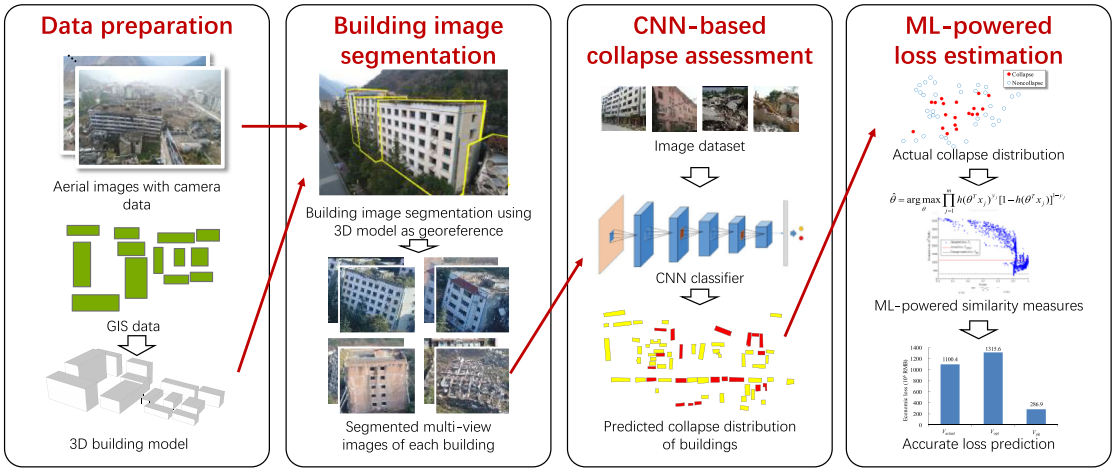



**Fig. 9.** Building seismic damage information acquisition based on drones and deep learning.

## 4. Disaster prevention and mitigation implementation based on the feedback loop from virtuality to reality

### *4.1. Pre-disaster planning and personnel evacuation drills*

Rational pre-disaster planning is of great importance for reducing disaster losses. The above disaster simulations based on digital twins can provide detailed damage predictions for buildings and cities, as well as high-fidelity disaster scenarios. These outcomes can help managers identify high-risk areas in cities more clearly, enabling more scientific and quantitative decision-making for reinforcement and renovation plans to enhance disaster resistance (Fig. 10) [56]. In addition, based on disaster simulations using digital twins, Xu et al. [41] utilized a virtual reality (VR) platform to conduct virtual drills for personnel rescue, providing a reference for facilitating evacuation and rescue efficiency during disasters (Fig. 11).

### *4.2. Near real-time post-disaster damage assessment and accident investigation*

Near real-time assessment of seismic damage can provide an important reference for decision-making regarding emergency rescue and post-earthquake recovery. Therefore, taking advantage of the dense strong earthquake monitoring network constructed in countries such as China, Lu et al. [57] developed a near real-time earthquake damage assessment system known as “RED-ACT” (that is, Real-time Earth-




**Fig. 10.** Visualization of building seismic damage in Tangshan City.

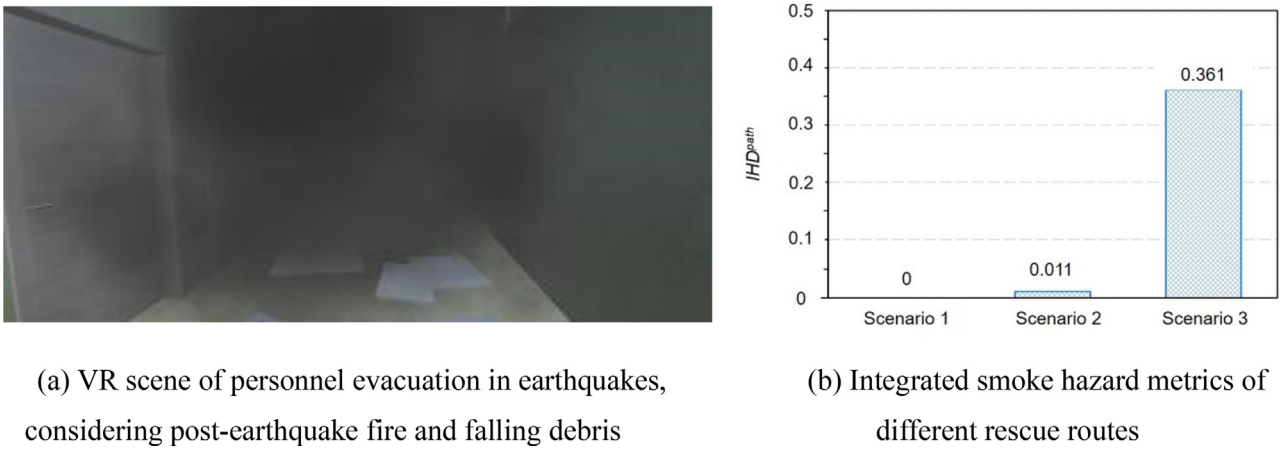



**Fig. 11.** VR scene of post-earthquake evacuation.

quake Damage Assessment using City-scale Time-history analysis) based on city-scale THA. Specifically, actual ground motion records obtained from seismic stations are input into the building models of the earthquake-stricken area, and a nonlinear THA of these models is subsequently performed using a high-performance computing platform. Seismic damage to buildings in the target region subjected to this earthquake can be evaluated based on the analysis results. The economic losses and repair times of the earthquake-stricken areas are calculated using the engineering demand parameters obtained from the THA. The system has been applied to hundreds of actual earthquakes in China and other countries, such as the United States and Japan, facilitating efficient emergency response and post-earthquake recovery (Fig. 12).

In addition, digital twin-based disaster simulations can be used for accident investigations to develop rational accident impact mitigation solutions. For instance, by combining a finite element model with a physics engine, Lu et al. [58] reproduced the collapse process and ruin distribution of a real condominium building collapse accident in Florida. On June 24, 2021, a 40-year-old reinforced concrete flat-plate structure building in Florida suffered a sudden partial collapse. The simulated collapse process and debris distribution agreed well with the actual conditions, indicating that the collapse initiated from the column removal zone and propagated to the east side of the building owing to the collision of the falling components (Fig. 13).

## 5. Case study

### 5.1. Digital twin-based investigation of the SEG tower

A super high-rise building in southern China, the SEG Tower, experienced an unusual vibration in May 2021, necessitating an urgent investigation into its cause and the provision of viable solutions. Consequently, the authors' research group employed a digital twin system framework to address this issue. A computer vision-based method was used to acquire the structural deformation information, enabling the capture of precise time-varying data pertaining to the vibrations occurring at the top of the building's mast. Subsequently, a comprehensive digital twin model was established (Fig. 14), encompassing the overall finite element model of the building, a refined finite element model of the mast, a model representing the surrounding wind field, and a fluid–solid coupling model for the mast. These numerical models facilitated fluid mechanics and structural dynamics analyses, leading to the identification of the root causes of the abnormal building vibrations, that is, vortex-induced mast vibration due to wind and the resonance of higher-order vibration modes within the primary structure of the building. By evaluating various remedial scenarios within the virtual model, it was determined that the removal of the mast would mitigate the occurrence of wind-induced vibrations in the building. These findings provide valuable guidelines for resolving such incidents.

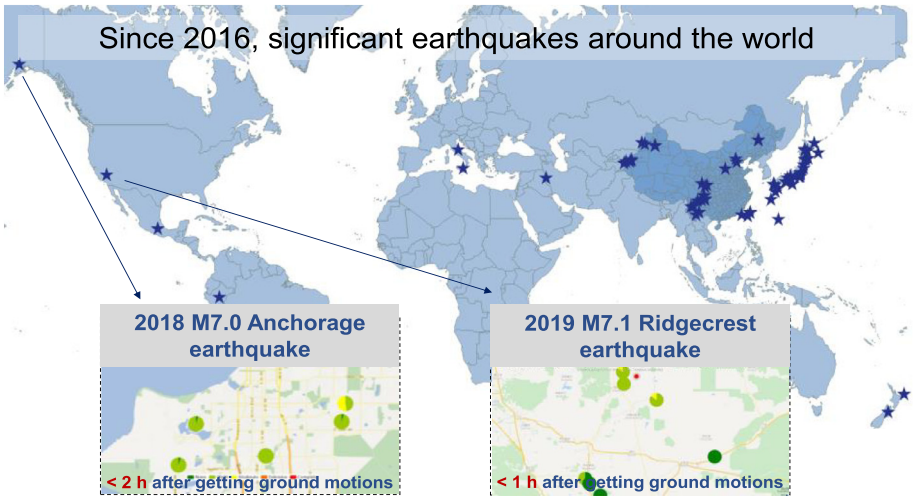



**Fig. 12.** Applications of RED-ACT.




**Fig. 13.** Comparison of simulated and actual collapse processes (photos via Andy Slater).

### *5.2. Digital twin platform for multi-hazard simulation of a campus*

Taking a university campus as a representative case study area, the authors' research group, along with the Central South Architectural Design Institute Co., Ltd., and the Dassault Company, collaboratively developed a multi-hazard simulation digital twin platform. This platform integrates data from 619 buildings, 19,740 trees, and essential lifeline systems such as water and electricity networks within the campus. It enables the realistic visualization of simulation outcomes for various disasters, including earthquakes, wind disasters, and fires, occurring within the vicinity of the campus. Consequently, the platform offers crucial technical support for

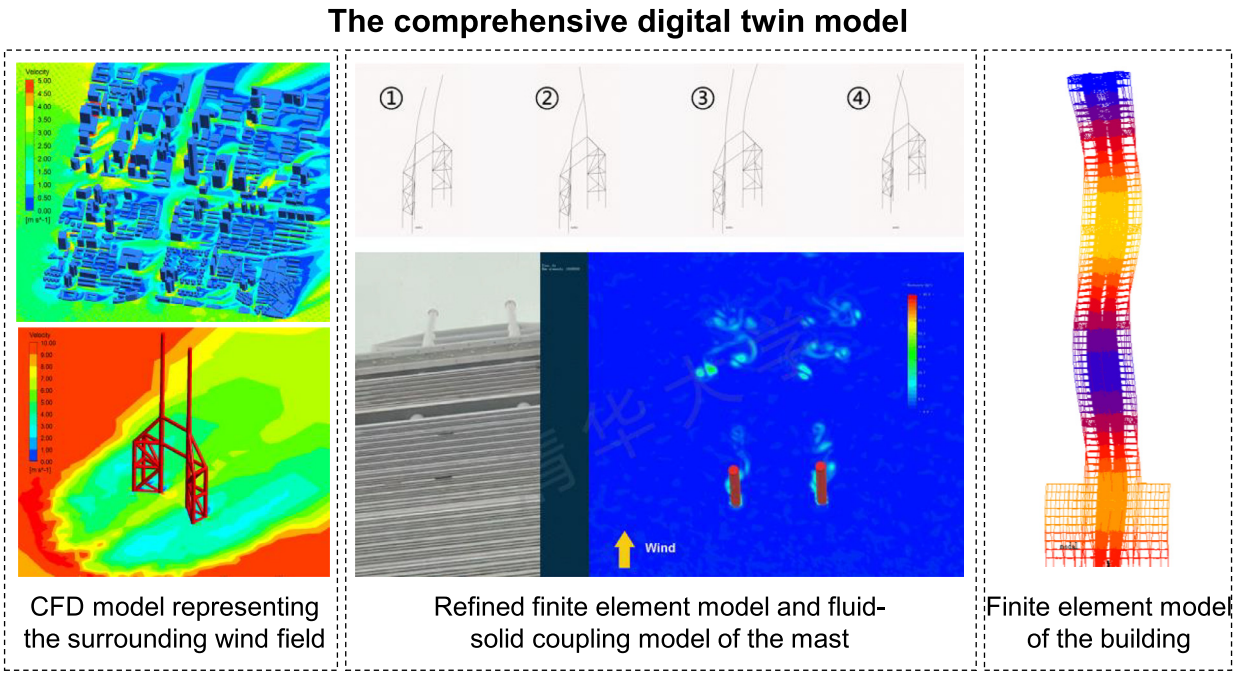



**Fig. 14.** Digital twin-based investigation of the SEG Tower.

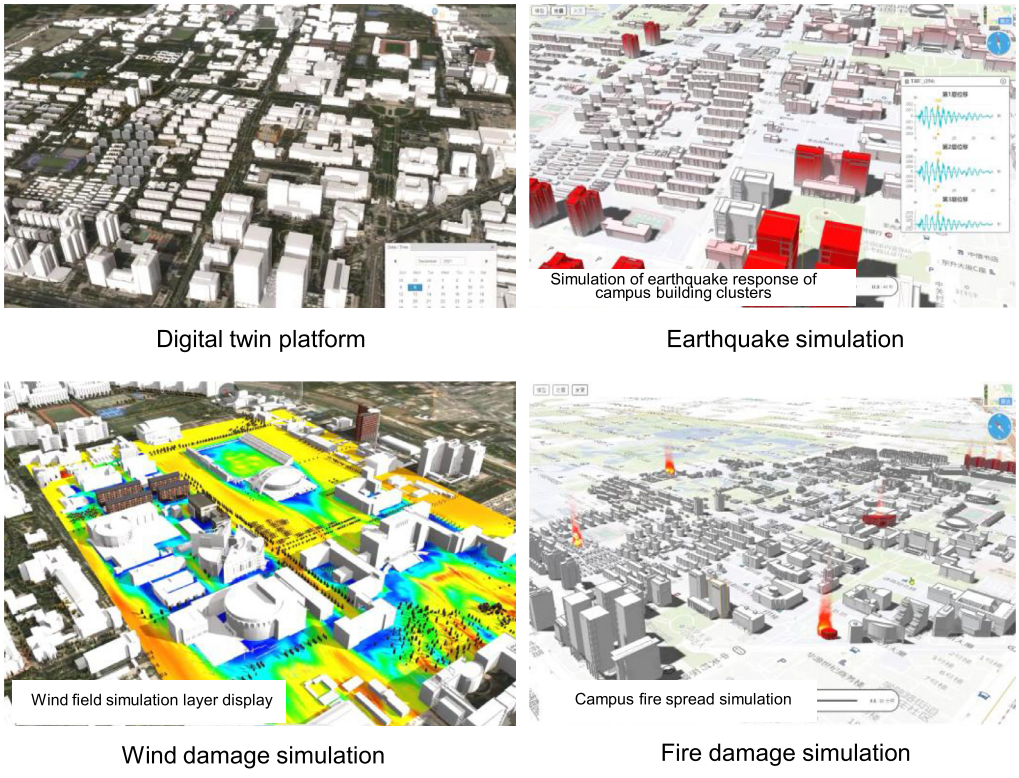



**Fig. 15.** Multi-hazard simulation digital twin platform for a campus.

pre-disaster planning and real-time assessment during emergencies (Fig. 15).

Wind-induced tree damage prediction is a typical application of the digital twin platform [59]. On April 15, 2021, a weather forecast predicted the imminent occurrence of a severe wind disaster in the campus area. The digital twin platform successfully predicted the distribution of potential damage to campus trees within half an hour. The observed post-disaster investigation of wind damage closely aligned with the predicted scenario (Fig. 16), exemplifying the prospects of this technology for effectively serving practical applications related to disaster prevention and mitigation. In addition to tree damage, the digital twin method can also be used to evaluate the occupant comfort caused by wind-induced responses of building clusters under different levels of strong winds [60].




**Fig. 16.** Comparison of the wind-induce tree damage prediction result and actual damage on the campus.

## 6. Conclusion

The construction of disaster-resilient cities has attracted considerable attention. This paper focuses on three key aspects pertaining to the integration of the digital twin concept in research of building and city resilience. These aspects include the acquisition of data from reality to the virtual space, disaster simulation within the virtual space, and the translation of virtual insights into effective disaster prevention strategies in reality. This paper provides a comprehensive summary of the relevant research endeavors of the authors' research group and showcases their application effectiveness and potential for development across various stages, including pre-disaster planning, real-time emergency assessments, post-disaster rescue operations, and accident investigation. This paper offers references for related research by presenting typical case studies.

Moreover, digital twinning methodologies rely significantly on comprehensive city inventory datasets. Additionally, the intricate nature of interdependencies among disasters and interactions within urban areas presents considerable challenges when simulating disaster scenarios in densely populated regions. These factors impose limitations on the practical implementation of digital twinning. Therefore, future research endeavors should prioritize the exploration of efficient and accurate methods for acquiring urban information data. This endeavor necessitates considering the interdependencies among different disasters to effectively conduct disaster scenario simulations for buildings and cities that face multiple hazards. Furthermore, there is a critical need to enhance the utilization of digital twinning in disaster prevention and mitigation strategies and to employ it as a tool for identifying potential additional disaster types.

## Relevance to resilience

This work addresses the critical applications of digital twin in research on building and urban resilience. The use of digital twin directly contributes to an increase in the resilience of buildings and cities, demonstrating significant effectiveness in pre-disaster planning, real-time emergency assessments, and other aspects.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## CRediT authorship contribution statement

**Yixing Wang:** Writing – original draft, Writing – review & editing. **Qingrui Yue:** Supervision. **Xinzheng Lu:** Supervision. **Donglian Gu:** Methodology, Software, Visualization, Writing – review & editing. **Zhen Xu:** Methodology. **Yuan Tian:** Methodology. **Shen Zhang:** Methodology.

## Acknowledgments

This work was supported by the National Natural Science Foundation of China (Nos. 52238011, 52208456), China National Postdoctoral Program for Innovative Talents (BX20220031), and Shenzhen Science and Technology Program (ZDSYS20210929115800001).

## References

[1] Acuto M, Parnell S, Seto KC. Building a global urban science. Nat Sustain 2018;1:2–4.
[2] Franco S. Sustainable cities and communities: the road towards SDG 11. TeMA-J Land Use, Mob Environ 2022;15(2):341–4.
[3] León JA, Ordaz M, Haddad E, et al. Risk caused by the propagation of earthquake losses through the economy. Nat Commun 2022;13(1):2908.
[4] Gitas IZ, Polychronaki A, Katagis T, et al. Contribution of remote sensing to disaster management activities: a case study of the large fires in the Peloponnese, Greece. Int J Remote Sens 2008;29(6):1847–53.
[5] Liu G, Li X, Wang J, et al. Research on the statistical characteristics of typhoon frequency. Ocean Eng 2020;209:107489.
[6] Nakashima M, Nagae T, Enokida R, et al. Experiences, accomplishments, lessons, and challenges of E-defense-Tests using world's largest shaking table. Jpn Archit Rev 2018;1(1):4–17.
[7] Del Carpio R M, Mosqueda G, Lignos DG. Experimental investigation of steel building gravity framing systems under strong earthquake shaking. Soil Dyn Earthq Eng 2019;116:230–41.
[8] Hofmann A, Muehlnikel R. Experimental and numerical investigation of fire development in a real fire in a five-storey apartment building. Fire Mater 2011;35(7):453–62.
[9] Pylianidi C, Osinga S, Athanasiadis IN. Introducing digital twins to agriculture. Comput Electron Agric 2021;184:105942.
[10] Jiang Y, Yin S, Li K, Luo H, Kaynak O. Industrial applications of digital twins. Philos Trans R Soc A 2021;379(2207):20200360.
[11] Sharma A, Kosasih E, Zhang J, Brintrup A, Calinescu A. Digital twins: state of the art theory and practice, challenges, and open research questions. J Ind Inf Integr 2022:100383.
[12] Grieves MW. Product lifecycle management: the new paradigm for enterprises. Int J Prod Dev 2005;2(1–2):71–84.
[13] Tuegel E. The airframe digital twin: some challenges to realization. In: 53rd AIAA/ASME/ASCE/AHS/ASC structures, structural dynamics and materials conference. Hawaii: Aerospace Research Central; 2012. p. 1812.
[14] Glaessgen E, Stargel D. The digital twin paradigm for future NASA and U.S. Air Force vehicles. In: 53rd AIAA/ASME/ASCE/AHS/ASC structures, structural dynamics and materials conference. Hawaii: Aerospace Research Central; 2012. p. 1818.
[15] Tao F, Zhang H, Liu A, et al. Digital twin in industry: state-of-the-art. IEEE Trans Industr Inform 2018;15(4):2405–15.

[16] Liu M, Fang SL, Dong HY, et al. Review of digital twin about concepts, technologies, and industrial applications. J Manuf Syst 2021;58:346.
[17] Lo CK, Chen CH, Zhong RY. A review of digital twin in product design and development. Adv Eng Inform 2021;48:101297.
[18] Tao F, Sui FY, Liu A, et al. Digital twin-driven product design framework. Int J Prod Res 2019;57(12):3935.
[19] Tao F, Liu A, Hu TL, et al. Digital twin driven smart design. Amsterdam: Elsevier Academic Press; 2020. p. 205.
[20] Tao F, Cheng JF, Qi QL, et al. Digital twin-driven product design, manufacturing and service with big data. Int J Adv Manuf Technol 2018;94:3563.
[21] Zhou GH, Zhang C, Li Z, et al. Knowledge-driven digital twin manufacturing cell towards intelligent manufacturing. Int J Prod Res 2020;58(4):1034.
[22] Balta EC, Tilbury DM, Barton K. A digital twin framework for performance monitoring and anomaly detection in fused deposition modeling. In: 2019 IEEE 15th international conference on automation science and engineering (CASE). Vancouver: IEEE; 2019. p. 823.
[23] Qamsane Y, Phillips JR, Savaglio C, et al. Open process automation-and digital twin-based performance monitoring of a process manufacturing system. IEEE Access 2022;10:60823.
[24] Fera M, Greco A, Caterino M, et al. Towards digital twin implementation for assessing production line performance and balancing. Sensors 2019;20(1):97.
[25] Ariyachandra M, Brilakis I. Detection of railway masts in airborne LiDAR data. J Constr Eng Manag 2020;146(9):04020105.
[26] Xiong XH, Adan A, Akinci B, et al. Automatic creation of semantically rich 3D building models from laser scanner data. Autom Constr 2013;31:325.
[27] Fiorillo F, Jiménez Fernández-Palacios B, Remondion F, et al. 3D surveying and modelling of the archaeological area of paestum, Italy. Virtual Archaeol Rev 2015;4(8):55–60.
[28] Wu B, Xie LF, Hu H, et al. Integration of aerial oblique imagery and terrestrial imagery for optimized 3D modeling in urban areas. ISPRS J Photogramm Remote Sens 2018;139:119.
[29] Lu QC, Parlikad AK, Woodall P, et al. Developing a digital twin at building and city levels: case study of West Cambridge campus. J Manag Eng 2020;36(3):05020004.
[30] White G, Zink A, Codeca L, et al. A digital twin smart city for citizen feedback. Cities 2021;110:103064.
[31] Enderami SA, Sutley EJ, Mazumder RK, et al. Virtual testbeds for community resilience analysis: step-by-step development procedure and future orientation. Resil Cities Struct 2023;2(2):42–56.
[32] Enderami SA, Mazumder RK, Dumler M, et al. Virtual testbeds for community resilience analysis: state-of-the-art review, consensus study, and recommendations. Nat Haz Rev 2022;23(4):03122001.
[33] Ellingwood BR, Cutler H, Gardoni P, et al. The Centerville Virtual Community: a fully integrated decision model of interacting physical and social infrastructure systems. Sustain Resilient Infrastruct 2016;1(3–4):95–107.
[34] Deierlein G.G., Zsarnóczay A. State of the art in computational simulation for natural hazards engineering (Version v2). 2021, 2021–01.
[35] Deierlein GG, McKenna F, Zsarnóczay A, et al. A cloud-enabled application framework for simulating regional-scale impacts of natural hazards on the built environment [Methods]. Front Built Environ 2020;6(196):558706.
[36] Braik AM, Koliou M. A novel digital twin framework of electric power infrastructure systems subjected to hurricanes. Int J Dis Risk Red 2023;97:104020.
[37] Wang C, Yu Q, Law KH, et al. Machine learning-based regional scale intelligent modeling of building information for natural hazard risk management. Autom Constr 2021;122:103474.
[38] Attary N, van de Lindt JW, Mahmoud H, et al. Hindcasting community-level damage to the interdependent buildings and electric power network after the 2011 Joplin, Missouri, Tornado. Nat Hazards Rev 2019;20(1):04018027.
[39] Lu X, Lu XZ, Guan H, et al. Application of earthquake-induced collapse analysis in design optimization of a supertall building. Struct Design Tall Special Build 2016;25:926–46.
[40] Lu XZ, Yang ZB, Xu Z, et al. Scenario simulation of indoor post-earthquake fire rescue based on building information model and virtual reality. Adv Eng Softw 2020:143.
[41] Xu Z, Lu XZ, Guan H, et al. A virtual reality based fire training simulator with smoke hazard assessment capacity. Adv Eng Softw 2014;68:1–8.
[42] Lu XZ, Gu DL, Xu Z, et al. CIM-Powered multi-hazard simulation framework covering both individual buildings and urban areas. Sustainability 2020;12(12):5059.
[43] Tian Y, Lin KQ, Lu XZ, Zhang L, Li Y, Guan H. Experimental and theoretical study of seismic and progressive collapse resilient composite frames. Soil Dyn Earthq Eng 2020;139:106370.
[44] Lu XZ, Guan H. Earthquake disaster simulation of civil infrastructures: from tall buildings to urban areas. Singapore: Springer; 2021. p. 21.
[45] Xu Z, Xue QR, Lu XZ, et al. 3-D fire-spreading model for building clusters with large ground elevation variations. J Tsinghua Univ (Sci Technol) 2020;60(1):95–100.
[46] Lu XZ, Zeng X, Xu Z, et al. Physics-based simulation and high-fidelity visualization of fire following earthquake considering building seismic damage. J Earthq Eng 2017;23(7):1173–93.
[47] Gu DL, Shuai QW, Wang YX. CIM-powered physics-based assessment of wind damages to building clusters considering trees. Dev Built Environ 2023:100178.
[48] Xu YJ, Lu XZ, Cetiner B, et al. Real-time regional seismic damage assessment framework based on long short-term memory neural network. Comput Aided Civil Infrastruct Eng 2021;36:504–21.
[49] Lu XZ, Xu YJ, Tian Y, et al. A deep learning approach to rapid regional post-event seismic damage assessment using time-frequency distributions of ground motions. Earthq Eng Struct Dyn 2021;50:1612–27.
[50] Sun CJ, Zhang F, Zhao PJ, et al. Automated simulation framework for urban wind environments based on aerial point clouds and deep learning. Remote Sens (Basel) 2021;13(12):2383.
[51] Gu DL, Wang C, Lu XZ. Automated assessment of wind damage to windows of buildings at a city scale based on oblique photography deep learning and CFD. J Build Eng 2022;52:117–34.
[52] Sun CJ, Gu DL, Zhang Y, et al. Vision-based displacement measurement enhanced by super-resolution using generative adversarial networks. Struct Control Health Monit 2022;29(10):3048.
[53] Sun CJ, Gu DL, Lu XZ. Three-dimensional structural displacement measurement using monocular vision and deep learning based pose estimation. Mech Syst Signal Process 2023;190:110141.
[54] Xiong C, Li QS, Lu XZ. Automated regional seismic damage assessment of buildings using an unmanned aerial vehicle and a convolutional neural network. Autom Constr 2020;109:102994.
[55] Lu XZ, Zeng X, Xu Z, et al. Improving the accuracy of near real-time seismic loss estimation using post-earthquake remote sensing images. Earthq Spectra 2018;34(3):1219–45.
[56] Cheng QL, Xu Z, Lu XZ, et al. Predicting the seismic damage of buildings in Tangshan city using the urban dynamic elasto-plastic analysis method. J Nat Dis 2018;27(1):71–80.
[57] Lu XZ, Cheng QL, Xu Z, et al. Real-time city-scale time-history analysis and its application in resilience-oriented earthquake emergency response. Appl Sci 2019;9(17):3497.
[58] Lu XZ, Guan H, Sun HL, et al. A preliminary analysis and discussion of the condominium building collapse in surfside, Florida, US, June 24, 2021. Front Struct Civil Eng 2021;15:1097–110.
[59] Gu DL, Zhao PJ, Chen W, Huang YL, Lu XZ. Near real-time prediction of wind-induced tree damage at a city scale: simulation framework and case study for Tsinghua University campus. Int J Disaster Risk Reduct 2021;53:102003.
[60] Gu DL, Kareem A, Lu XZ, Cheng QL. A computational framework for the simulation of wind effects on buildings in a cityscape. J Wind Eng Ind Aerodyn 2023;234:105347.